Dohyun (Eric) Kim

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EDUCATION

University of Auckland — **Auckland, New Zealand**
Bachelor of Engineering (Honours) in Computer Systems Engineering — *Expected Graduation Nov 2027*

- **Concentrations**: Embedded Systems & Hardware Design
- **Related Courseworks**: Electrical Engineering ▪ Electronics ▪ Electromagnetics ▪ Signals & Control Systems ▪ AI & Machine Learning ▪ Computer Architecture ▪ Hardware-Software Systems ▪ Digital Systems Design

EXPERIENCES

Research Assistant — **Auckland, New Zealand**
University of Auckland — *Jul 2026 - Present*

- Contribute to the **robot soccer system** at UoA's robotics lab (CARES), including **control and decision making logic**, with future work migrating its classical architecture into a modern software framework.
- Develop and test navigation algorithms for **TurtleBot2**, with sensor integration and performance evaluation.

Robotics Instructor — **Auckland, New Zealand**
Creative Imaginary Lab (ciLab) — *Apr 2026 - Present*

- Instructed **50+ students** in robot **hardware assembly and software programming** to complete missions.
- Supported student development and coached teams in preparation for **nationwide robotics competitions**.

SELECTED PROJECTS

Flappy Universe — **VHDL & FPGA**
FPGA Game Implementation - Team Project — *Jun 2026*

- Implemented a Flappy Bird style game in **VHDL**, using a custom VGA sync generator and PLL based clock division on a **DE0 CV FPGA**, **to render real time sprite based gameplay.**
- Built a **hardware LFSR** random number generator, using **shift register feedback logic**, to procedurally place pipe gaps and drive collision detection and scoring.

Smart Energy Monitor — **Embedded C & PCB**
Embedded Energy Monitoring Systems - Team Project — *Oct 2025*

- Designed a dual channel energy monitoring system, using **signal conditioned ADC sampling on an ATmega328P**, to compute RMS voltage, peak current, and average power in real time.
- Built and validated a double layer PCB, using **Altium Designer** and UART based output, **to meet a ±5% full scale accuracy spec** across a 2.5 to 7.5 VA load range.

AWARDED PROJECT

3rd Place in ECSE Design Competition 2025 — **Team Competition**
"Winnie the Bot" - Interactive companion robot — *Sep 2025*

- Contributed to an AI powered interviewer robot, using **dual ATmega328P Nano** microcontrollers **to drive face tracking, arm movement, and real time voice interaction.**
- **Prototyped and refined the robot's enclosure**, using iterative 3D printing and multiple design revisions, to deliver a compact, durable, and functional physical build.

LEADERSHIPS

Academic Team Executive — **Auckland, New Zealand**
KEB (Korean Engineering Body) - Founding Member — *Jul 2024 - Present*

- Developed and delivered tutorial sessions for **20+ junior engineering students** across various courses.
- Participated in the planning and delivery of academic events **to support the student community**.

Official Competition Staff — **NZ Robotics Olympiad 2026**
IEEE R&A (IEEE Robotics and Automation Society) — *Jul 2026*

- **Selected as official Competition Staff** based on prior volunteering & robotics instructor experience at ciLab.
- Led competition operations and **resolved technical issues**, ensuring smooth running matches for all teams.

Full-time Student Volunteer — **NZ Robotics Olympiad 2025**
IEEE (Institute of Electrical and Electronics Engineers) — *Jul 2025*

- **Volunteered 40+ hours** supporting operations and logistics at NZRO 2025, an IEEE robotics competition.
- Collaborated with organisers **to ensure a smooth event experience** for participants.

SKILLS

Languages & Platforms: C, VHDL, Python, Java, R, MATLAB, ROS, Git, AWS
Hardware & EDA Tools: Altium Designer, LTSpice, ModelSim, Intel Quartus Prime, Proteus, Atmel AVR, AutoCAD